Magnetic Levitation Train Control System Design

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***Abstract*— Controlling a multi-carriage maglev train presents a fundamental challenge: only the lead carriage is actuated, yet all carriages must reach cruise speed comfortably and without mechanical divergence while relying on noisy velocity measurements alone. This paper presents a discrete-time Linear Quadratic Regulator (LQR) with integral action and a Kalman filter addressing these requirements. The nonlinear four-state model is reduced to three observable states, linearised at $70\,\mathbf{m/s}$ to extend performance across the full acceleration range, and discretised for digital implementation. A hard $1\,\mathbf{m/s^2}$ output clamp enforces passenger comfort, with conditional anti-windup preventing integrator windup during saturation. The Kalman filter suppresses sensor noise without amplifying it into the control signal, and integral action guarantees zero steady-state error by the internal model principle. Simulation on the full nonlinear plant confirms all objectives: both carriages reach $100\,\mathbf{m/s}$ within $45\,\mathbf{s}$, peak acceleration is $0.5\,\mathbf{m/s^2}$, and peak coupler deflection is $0.24\,\mathbf{m}$. The main limitations suggest that gain scheduling and a Gauss–Markov bias-state augmentation should be implemented.**

I. Introduction

Magnetic levitation trains eliminate wheel-to-rail contact, reducing friction and enabling operational speeds far exceeding those of conventional rail. As maglev networks expand globally, the control of multi-carriage configurations becomes increasingly important: each carriage contributes distinct aerodynamic drag and inertia, and only the leading carriage is typically actuated, coupling all carriages through mechanical connectors.

The system considered here is a two-carriage maglev train in which thrust u is applied solely to carriage 1 (mass M_1). Carriage 2 (mass M_2) responds passively through a nonlinear spring-damper coupler. Three simultaneous objectives must be met: (i) drive and maintain both carriages at $100\,\mathrm{m/s}$; (ii) limit carriage acceleration to $1\,\mathrm{m/s^2}$ for passenger comfort; and (iii) prevent inter-carriage spacing from diverging from the coupler's natural length. Quadratic aerodynamic drag and coupler nonlinearity complicate the design, and since only carriage velocities are measurable, all remaining states must be estimated.

Section II develops the nonlinear model, performs linearisation, and resolves a structural unobservability through state reduction. Section III presents the discrete-time LQR, Kalman filter, integral action, and saturation handling. Section V evaluates closed-loop performance on the full nonlinear plant.

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II. System Modelling and Linearisation

A. Nonlinear Equations of Motion

The dynamics of the coupled two-carriage system are

$$\dot{x}_1 = x_2, \tag{1}$$
$$M_1\dot{x}_2 = u - D_{a1}(x_2) - F_c(\delta_x,\delta_v), \tag{2}$$
$$\dot{x}_3 = x_4, \tag{3}$$
$$M_2\dot{x}_4 = -D_{a2}(x_4) + F_c(\delta_x,\delta_v), \tag{4}$$

where x_1, x_3 are positions and x_2, x_4 are velocities of carriages 1 and 2, $\delta_x = x_1 - x_3 - L_c$ is the coupler deformation, and $\delta_v = x_2 - x_4$ is the relative velocity. Negligible rail contact means drag is purely aerodynamic,

$$D_{ai}(x_{2i}) = b_{ai}\, x_{2i}|x_{2i}|, \tag{5}$$

and the coupler force combines linear and nonlinear stiffness with linear damping,

$$F_c(\delta_x,\delta_v) = k_c\,\delta_x + d_c\,\delta_v + k_{nl}\,\delta_x^3, \tag{6}$$

where k_c, d_c, and k_{nl} are the linear spring, linear damping, and nonlinear spring constants, and L_c is the coupler natural length. Parameter values are listed in Table I. Only carriage velocities are measurable: $y_1 = x_2$, $y_2 = x_4$.

TABLE I
System Parameters

Parameter	**Carriage 1**	**Carriage 2**
Mass M_i (kg)	9000	10000
Drag coeff. b_{ai} ($\mathrm{N\,s^2/m^2}$)	0.50	0.45
Linear spring k_c (N/m)	5×10^4	
Linear damping d_c (N s/m)	1×10^5	
Nonlinear spring k_{nl} ($\mathrm{N/m^3}$)	1×10^4	
Coupler natural length L_c (m)	10	

B. Controllability and State Reduction

Evaluating the controllability matrix $\mathcal{C} = \begin{bmatrix} B & AB & A^2B & A^3B\end{bmatrix}$ at a representative operating point gives $\mathrm{rank}(\mathcal{C}) = 4$, confirming the full system is controllable. However, the observability matrix $\mathcal{O} = \begin{bmatrix} C^\top & (CA)^\top & (CA^2)^\top & (CA^3)^\top\end{bmatrix}^\top$ has rank 3, indicating the four-state system is *not* fully observable from velocity measurements alone. Physically, the two absolute positions x_1 and x_3 are indistinguishable through velocity sensing; only their difference carries observational content.

The state vector is therefore reduced to

$$\mathbf{z} = \begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \end{bmatrix} = \begin{bmatrix} x_1 - x_3 \\ x_2 \\ x_4 \end{bmatrix}, \tag{7}$$

replacing the two absolute positions with their relative displacement. The observability matrix of the reduced system (A_r, C_r) has full rank 3, confirming all states—including the unmeasured coupler deformation Z_1—are reconstructible from velocity measurements. Controllability of the reduced system is also verified, preserving the original rank-4 property in reduced form.

C. Linearisation

The reduced model is linearised about the cruise equilibrium at $v^* = 100\,\text{m/s}$. At this point $\dot{x}_2 = \dot{x}_4 = 0$, $x_2 = x_4 = v^*$, and $\delta_x = 0$, yielding the equilibrium thrust

$$u^* = (b_{a1} + b_{a2})(v^*)^2 = 9500\,\text{N}, \tag{8}$$

This model simplification assumes the coupler is 0 deflection at equilibrium and therefore applies no force. In reality, there is a small static deflection as it is carrying carriage two's drag but this was found to be inconsequential to the simulation.

To improve closed-loop performance during acceleration, the drag Jacobian terms are evaluated at $v_{\text{lin}} = 70\,\text{m/s}$ rather than v^*. The system consistently performed well at cruise speed regardless of linearisation point, but was sluggish at lower speeds when linearised at $v^* = 100\,\text{m/s}$; evaluating the Jacobian at the lower speed produces a more conservative model that extends effective performance across the full acceleration range. Computing the Jacobian of (1)–(4) in reduced coordinates yields

$$A_r = \begin{bmatrix} 0 & 1 & -1 \\ \dfrac{-k_c}{M_1} & \dfrac{-(2b_{a1}v_{\text{lin}} + d_c)}{M_1} & \dfrac{d_c}{M_1} \\ \dfrac{k_c}{M_2} & \dfrac{d_c}{M_2} & \dfrac{-(2b_{a2}v_{\text{lin}} + d_c)}{M_2} \end{bmatrix}, \tag{9}$$

which evaluates numerically to

$$A_r \approx \begin{bmatrix} 0.000 & 1.000 & -1.000 \\ -5.556 & -11.119 & 11.111 \\ 5.000 & 10.000 & -10.006 \end{bmatrix}, \tag{10}$$

with $B_r = \begin{bmatrix} 0 & M_1^{-1} & 0 \end{bmatrix}^\top$ and $C_r = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

III. Control and Observer Design

A. Discrete-Time Representation

The continuous-time eigenvalues of A_r are $\{-20.61,\ -0.51,\ -0.007\}$ rad/s; the fastest mode at -20.61 rad/s sets the sampling requirement. At $T_s = 0.01\,\text{s}$ the controller samples at $100\,\text{Hz}$, approximately $30\times$ the fastest plant mode, consistent with the standard guideline of 20–40 times the bandwidth [2].

The continuous-time model (A_r, B_r, C_r) is converted to discrete time using the zero-order hold method,

$$\mathbf{z}[k+1] = G\,\mathbf{z}[k] + H\,\Delta u[k], \quad \mathbf{y}[k] = C_r\mathbf{z}[k], \tag{11}$$

where $G = e^{A_r T_s}$, $H = \int_0^{T_s} e^{A_r\tau}\mathrm{d}\tau\, B_r$, and $\Delta u = u - u^*$ is the perturbation input about equilibrium. Controllability and observability of (G, H, C_r) are verified numerically prior to design.

B. LQR Design

The discrete-time LQR minimises the infinite-horizon cost

$$J = \sum_{k=0}^{\infty} \big(\mathbf{z}[k]^\top Q\,\mathbf{z}[k] + \Delta u[k]^\top R\,\Delta u[k]\big), \tag{12}$$

yielding gain K such that $\Delta u[k] = -K\mathbf{z}[k]$.

Weighting matrices were initialised using Bryson's rule [1], setting $Q_{ii} = 1/\bar{z}_i^2$ and $R = 1/\bar{u}^2$ for maximum tolerable deviations $\bar{z}_i$ and $\bar{u}$. The weights were then refined to apply greater emphasis on coupler deflection than on velocity error. The final matrices are

$$Q = \mathrm{diag}(4000,\ 9,\ 9), \quad R = 5.49 \times 10^{-6}, \tag{13}$$

corresponding to a maximum tolerable coupler deformation of $0.5\,\text{m}$ ($Q_{11} = 4000$) and a maximum velocity error of $0.33\,\text{m/s}$ for each carriage ($Q_{22} = Q_{33} = 9$). The large Z_1 weight prioritises keeping the carriages from diverging. The resulting gain is:

$$K = \begin{bmatrix} 3210.3 & 948.4 & 729.0 \end{bmatrix}. \tag{14}$$

The closed-loop controller poles are $\{0.9991,\ 0.9947,\ 0.8138\}$, all within the unit circle, confirming discrete-time stability.

C. Integral Action and Anti-Windup

To eliminate steady-state velocity error due to aerodynamic drag, an integrator accumulates the carriage 1 velocity tracking error $e[k] = v^* - Z_2[k]$:

$$x_I[k] = x_I[k-1] + K_I\,e[k], \quad K_I = 200. \tag{15}$$

The integral output is clamped to $\pm 1140\,\text{N}$ ($12\,\%$ of u^*) to limit its contribution and prevent excessive windup during the acceleration phase. The total control input applied to the plant is

$$u[k] = \mathrm{clip}\big(\Delta u[k] + x_I[k] + u^*,\ 0,\ u_{\max}\big), \tag{16}$$

where $u_{\max} = M_1 a_{\max} + u^*$ with the passenger comfort limit $a_{\max} = 1\,\text{m/s}^2$, and $u_{\min} = 0$ (the train cannot produce reverse thrust). When the output saturates, the integral state is frozen:

$$x_I[k] = \begin{cases} x_I[k-1] + K_I e[k] & \text{if } 0 < u[k] < u_{\max}, \\ x_I[k-1] & \text{otherwise}, \end{cases} \tag{17}$$

preventing integrator wind-up and the associated overshoot on recovery. This integrator guarantees zero steady state error.

D. Kalman Filter and Observer Pole Trade-off

A discrete-time Kalman filter provides state estimates from noisy velocity measurements:

$$\hat{\mathbf{z}}[k+1] = G\hat{\mathbf{z}}[k] + H\Delta u[k] + L\big(\mathbf{y}[k] - C_r\hat{\mathbf{z}}[k]\big), \tag{18}$$

where the steady-state gain L is computed from the process noise covariance Q_{kf} and measurement noise covariance R_{kf}.

The covariances were tuned to address the observed noise characteristics: the diagonal entries of R_{kf} (0.01) are approximately 200 times larger than those of Q_{kf} (5×10^{-5}). This makes the filter trust model predictions substantially more than sensor readings. This was done after determining the system had a well characterised linear model relative to the significant measurement noise. The resulting Kalman gain is

$$L = \begin{bmatrix} -0.0414 & 0.0392 \\ 0.0502 & 0.0182 \\ 0.0182 & 0.0501 \end{bmatrix}. \tag{19}$$

The resulting observer poles are $\{0.9316,\ 0.9759,\ 0.8009\}$. While each is strictly closer to the origin than the corresponding controller pole—satisfying the separation principle—the margin is modest compared to the conventional recommendation of observer poles 2–5 times faster than the controller. This was an accepted design trade-off: placing observer poles further from the unit circle requires larger Kalman gains, which causes the filter to weight noisy sensor measurements more aggressively, producing noisier state estimates that propagate directly into the LQR control signal. Given the significant sensor noise present in this system, tighter noise suppression through a conservative $R_{\mathrm{kf}}/Q_{\mathrm{kf}}$ ratio was prioritised over aggressive pole separation. The resulting observer converges sufficiently before the control action drives the plant to cruise speed, and closed-loop stability is confirmed by simulation on the nonlinear plant.

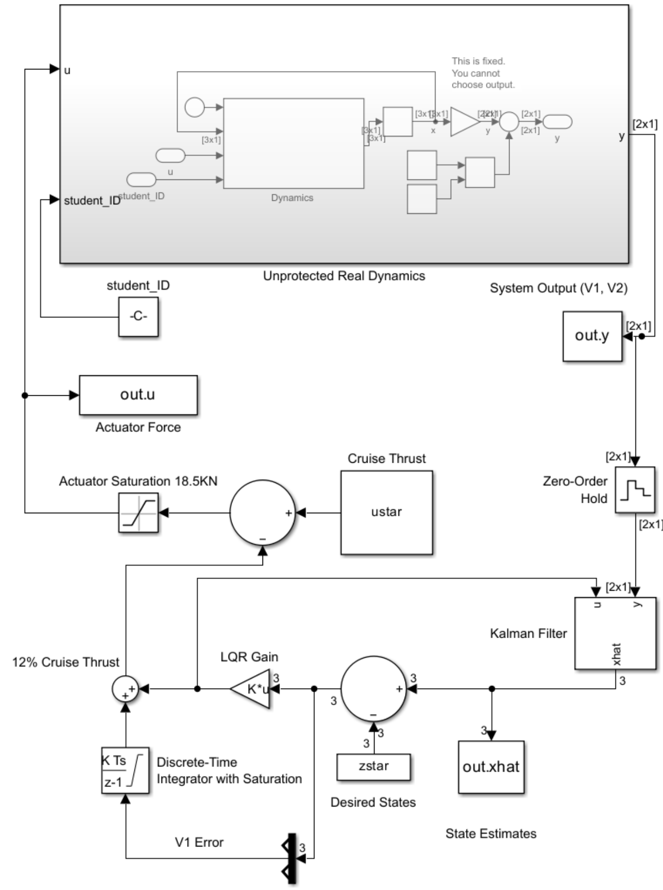

Fig. 1. Full Simulink Model

IV. Model Implementation In Simulink

The control system was developed and simulated using Simulink. The complete block diagram can be seen in Figure 1.

V. Results and Discussion

The closed-loop system was evaluated on the full nonlinear plant (1)–(4) over a 120 s simulation. The initial condition represents both carriages travelling 10 m/s below cruise speed ($x_2 = x_4 = 90$ m/s) with a coupler pre-deformation of $\delta_x(0) = 0.1$ m.

A. Velocity Tracking

Figure 2 shows the measured and Kalman-estimated speeds of both carriages against the cruise reference $v^* = 100$ m/s. Both carriages reach cruise speed within 45 s. Carriage 1, directly actuated, tracks the reference throughout; carriage 2 follows via coupler coupling with a transient lag consistent with its passive dynamics. Integral action eliminates residual steady-state error, and no overshoot is observed. The Kalman-estimated velocities closely follow the measured signals in trend while substantially reducing sensor noise fluctuations because it was tuned to trust its internal model more than the sensor readings. The smooth estimates prevent noise from exciting unnecessary control effort through the LQR law.

B. Control Input

Figure 3 shows the thrust profile $u[k]$ throughout the manoeuvre. The output remains within the clamped limits at all times, rising sharply at initialisation to accelerate the train and decaying smoothly toward the equilibrium value of $u^* = 9500$ N as cruise speed is approached. The anti-windup mechanism prevents integrator over-accumulation during the initial saturation phase, yielding clean monotonic convergence without oscillation on recovery.

C. Carriage Accelerations

The carriage accelerations derived from Kalman-filtered velocity estimates are shown in Figure 4. Carriage 1 shows a brief downward spike of roughly $-2\,\mathrm{m/s^2}$ in the first second, which is a filter-convergence artifact rather than a physical acceleration: the Kalman estimate begins with non-zero initial error, and differentiating the rapidly correcting estimate produces a transient that does not appear in the underlying plant motion. Once the observer settles, the peak sustained acceleration is $0.5\,\mathrm{m/s^2}$ for both carriages, well within the $1\,\mathrm{m/s^2}$ comfort constraint. The output clamping is therefore active but not at its hard limit, indicating the LQR gain naturally produces conservative thrust profiles. This is consistent with the low R weighting, which permits large control effort, combined with the Q_{11} coupler penalty that

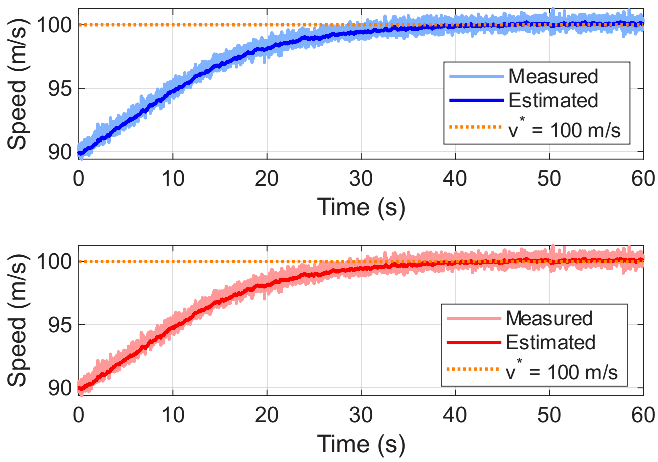

Fig. 2. Carriage speeds versus cruise reference $v^* = 100\,\mathrm{m/s}$. Top: carriage 1; bottom: carriage 2. Light traces: raw measurement; bold traces: Kalman estimate; dotted: v^*.

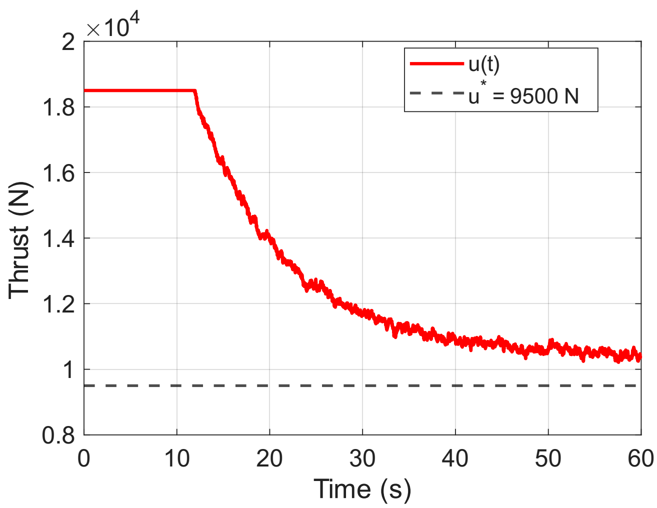

Fig. 3. Control input $u[k]$ (red) and equilibrium thrust $u^* = 9500\,\mathrm{N}$ required to maintain cruise speed (green dashed).

indirectly limits how aggressively the velocity error can be driven down. Accelerations decay smoothly to zero as both carriages reach cruise speed.

D. Coupler Deflection

Figure 5 shows the estimated coupler deformation $\hat{\delta}_x = \hat{Z}_1 - L_c$ over the simulation. From the initial pre-deformation of $0.1\,\mathrm{m}$, the estimate peaks at $0.24\,\mathrm{m}$ as the thrust transient loads the coupler, then decays to $0.022\,\mathrm{m}$ by $60\,\mathrm{s}$. A non-zero equilibrium deflection is physically required: both carriages hold cruise speed, so the coupler must transmit carriage 2's full aerodynamic drag of $b_{a2}(v^*)^2 = 4500\,\mathrm{N}$. Through the linear stiffness alone this requires a true deflection of $4500/k_c \approx 0.09\,\mathrm{m}$; the nonlinear term $k_{nl}\delta_x^3$ contributes less than $8\,\mathrm{N}$ at this deflection and is negligible. The estimate of $0.022\,\mathrm{m}$ therefore sits $\sim 0.07\,\mathrm{m}$ below the physically required value. The observer is linearised about $\delta_x = 0$, where the coupler carries no load, so it has no model of the static offset and reconstructs $\hat{Z}_1$ low. This bias is discussed

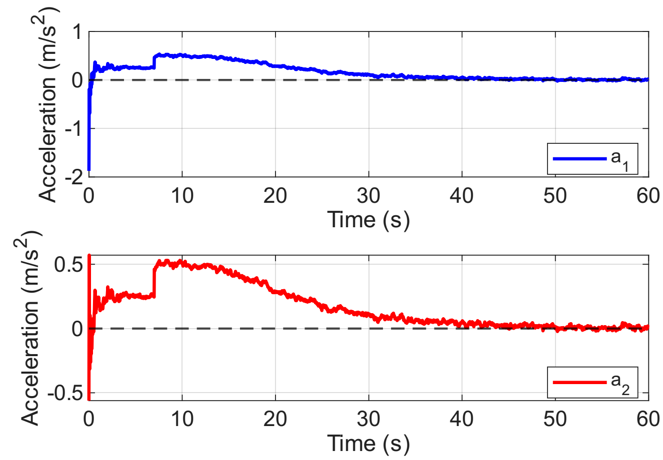

Fig. 4. Carriage accelerations derived from Kalman-filtered velocity estimates. Top: carriage 1; bottom: carriage 2. Dashed line at zero acceleration.

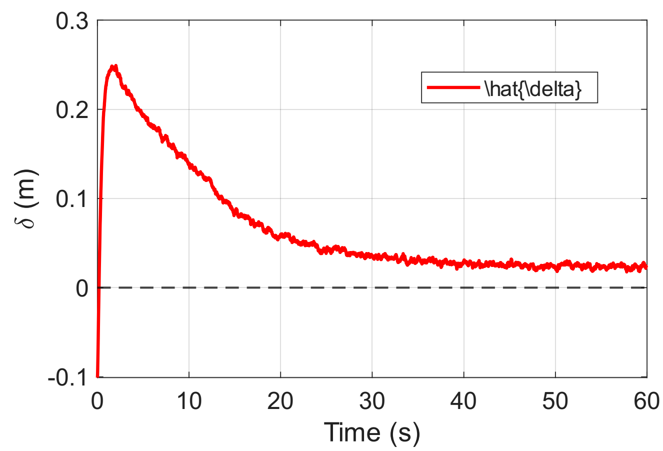

Fig. 5. Kalman-estimated coupler deformation $\hat{\delta}_x = \hat{Z}_1 - L_c$ (m). Dashed line at zero (coupler at natural length).

in the Model Limitations section, where a Gauss–Markov bias state is proposed as a correction. The peak estimate of $0.24\,\mathrm{m}$ remains well within the $0.5\,\mathrm{m}$ design tolerance, and the negligible nonlinear contribution at these deflections confirms the linearisation was justified.

VI. Model Limitations

The principal limitation is the single linearisation point: the LQR gain matrix is only optimal in a neighbourhood of $v_{\mathrm{lin}} = 70\,\mathrm{m/s}$, so at low speeds the over-estimated aerodynamic drag produces a conservative feedforward that marginally over-thrusts during early acceleration. This mismatch only affects the transient; integral action drives velocity error to zero regardless of linearisation point. Should the train operate across a significantly wider speed envelope, the linearisation error could grow large enough to destabilise the closed loop, the natural failure mode of a fixed-gain controller pushed far off-design; gain scheduling across multiple operating points would resolve this. The linearisation also neglects $k_{nl}\delta_x^3$, which is negligible at observed deformation magnitudes but would become significant under emergency braking or large mechanical disturbances, where stiffness hardening causes the true plant to diverge from the linear model and risks coupler oscillation the controller is not designed to damp; an extended Kalman filter incorporating the nonlinear stiffness term would address this failure mode. A further limitation is that the Kalman filter has no mechanism to correct its own state estimate error. Any persistent bias between the observer model and the true plant propagates into the estimate indefinitely, manifesting as the gap between the $\sim 0.09\,\mathrm{m}$ deflection physically required at cruise and the $0.022\,\mathrm{m}$ the filter reconstructs. Augmenting the observer with a Gauss–Markov bias state would provide an explicit correction mechanism for this offset.

VII. Conclusion

A discrete-time LQR with integral action and a Kalman filter was designed for a two-carriage maglev train. The four-state system is not fully observable from velocity measurements, so it was reduced to three states using the relative inter-carriage position and the two carriage velocities. The LQR weights came from Bryson's rule and were then adjusted to penalise coupler deflection more heavily, and the model was linearised at 70 m/s to balance the transient response at lower velocities with the steady state disturbance rejection at cruise speed. The Kalman filter was tuned to trust the model over the noisy measurements; this kept the observer poles only modestly faster than the controller, but that was preferred to the noisier estimates that higher Kalman gains would have fed into the control signal. Output clamping and integrator anti-windup held the train within the 1 m/s^2 comfort limit throughout. On the full nonlinear plant the design met every objective: both carriages reached 100 m/s within 45 s with no overshoot, peak acceleration stayed at 0.5 m/s^2, and the coupler deflection peaked at 0.24 m. The deflection estimate settled at 0.022 m, below the 0.09 m true equilibrium. This gap is an observer bias that the Gauss–Markov augmentation in Section VI would correct.

Acknowledgements

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References

[1] A. E. Bryson and Y. C. Ho, *Applied Optimal Control*. Hemisphere, 1975.

[2] G. F. Franklin, J. D. Powell and M. L. Emami-Naeini, *Feedback Control of Dynamic Systems*, 4th ed. Prentice Hall, 1998.